FIITJEE

ALL INDIA TEST SERIES

OPEN TEST

JEE (Main)-2020

TEST DATE: 09-02-2020

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. B

Sol. $\vec{v}_i = (v\cos\alpha)\hat{i} + (v\sin\alpha - gt)\hat{j}$

$< \vec{v} >= (v\cos\alpha)\hat{i} + (v\sin\alpha - \frac{gt}{2})\hat{j}$

$|\vec{v}_i| = |< \vec{v} >|$

$\Rightarrow t = \frac{4v\sin\alpha}{3g}$

2. B

Sol. $r = a(1-\cos\theta) \Rightarrow dr = a\sin\theta\, d\theta$

Flux through the element is

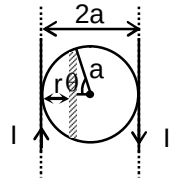
$$d\phi = \frac{\mu_0 I}{2\pi a}\left[\frac{1}{(1-\cos\theta)} + \frac{1}{(1+\cos\theta)}\right] 2a^2\sin^2\theta d\theta$$

$$\phi = \frac{\mu_0 I}{\pi}\int_0^{\pi}\frac{2a}{\sin^2\theta}\sin^2\theta d\theta = \frac{\mu_0 I}{\pi} 2a\int_0^{\pi} d\theta$$

$\phi = 2\mu_0 Ia$

$\therefore$ mutual inductance

$M = \phi/I = 2\mu_0 a$



3. B

Sol. $V = \frac{Q}{k4\pi\varepsilon_0 r}$

$$\Rightarrow \frac{Q}{k\varepsilon_0} = 4\pi rv \quad \ldots(i)$$

$$I = \oint_S \vec{J}\cdot d\vec{s} = \sigma \oint_S \vec{E}\cdot d\vec{s} = \frac{\sigma Q}{k4\pi\varepsilon_0 r^2}4\pi r^2 = \frac{\sigma Q}{k\varepsilon_0} = 4\pi\sigma rv$$

4. C

Sol. $\frac{q\lambda}{2\pi\varepsilon_0 r} - \frac{q\mu_0 I}{2\pi r}v = 0$

$$v = \frac{\lambda}{\varepsilon_0\mu_0 I} = \frac{\lambda c^2}{I}$$

5. A

Sol. Speed of sound, $v = \sqrt{\frac{B}{\rho}}$

$PV^2 = K$

Bulk modulus, $B = -V\frac{dP}{dV} = \frac{2K}{V^2} = 2P$

$$v = \sqrt{\frac{2P}{\rho}}$$

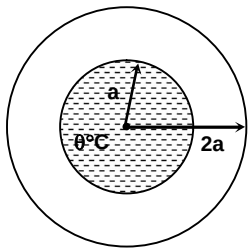
6. D

Sol. In steady state, potential difference across 6 μF capacitor = 7.5 V

$\therefore$ Charge on the 6μF capacitor, $q = 6 \times 7.5 = 45\ \mu C$

7. B

Sol. Thermal resistance of the spherical shell is



$$R = \int_a^{2a} \frac{dr}{K4\pi r^2} = \int_a^{2a} \frac{dr}{\frac{\alpha}{r^2}4\pi r^2} = \frac{a}{4\pi\alpha} \quad \ldots(i)$$

$$-\frac{msd\theta}{dt} = \frac{\theta - \theta_0}{R}$$

$$-mSR\int_{80}^{60} \frac{d\theta}{(\theta - \theta_0)} = \int_0^t dt$$

$$t = mSR\ell n\left(\frac{5}{3}\right)$$

$$t = \rho\frac{4}{3}\pi a^3 S\left(\frac{a}{4\pi\alpha}\right)\ell n\left(\frac{5}{3}\right)$$

$$t = \frac{\rho S a^4}{3\alpha}\ell n\left(\frac{5}{3}\right)$$

8. C

Sol. $A = A_0 e^{-\lambda t}$

$$7.8 = 31.2e^{-\frac{\ell n2}{5570}t}$$

$\Rightarrow -2\ell n2 = -\frac{\ell n2}{5570}t$

Solving, t = 11140 years

9. A

Sol. Let v = actual velocity of fly towards right

$v_{approach} = 4 + \mu v$

$8 = 4 + \mu v$

$\therefore \quad v = 3$ m/s

10. D

Sol. $T - mg = \frac{m\left(3v_0\right)^2}{\ell}$

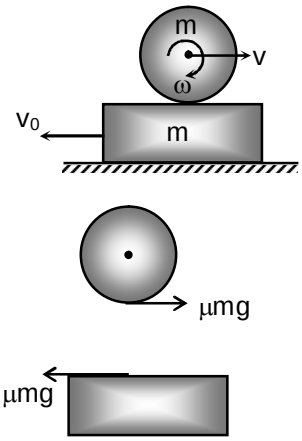
$T = mg + \frac{9mv_0^2}{\ell}$

11. A

Sol. $\tau_{CM} = I_{CM}\alpha$

$\mu mgR = \frac{mR^2}{2}\alpha$

$\alpha = \frac{2\mu g}{R}$



When pure rolling motion starts

$\omega R - v = v_0$

$(\omega_0 - \alpha t)R - \mu gt = \mu gt$

$\omega_0 R - 2\mu gt - \mu gt = \mu gt$

$\omega_0 R = 4\mu gt$

$\Rightarrow t = \frac{\omega_0 R}{4\mu g} = \frac{40 \times 0.4}{4 \times 0.2 \times 10} = 2\,\text{sec}$

12. A

Sol. The equation of process AB is

$P = \left(\frac{P_0}{V_0}\right)V \Rightarrow PV^{-1} = \frac{P_0}{V_0} = \text{constant}$

$\therefore$ This is a polytropic process with a polytropic constant, x = –1

$\therefore$ Molar heat capacity, $C = C_V + \frac{R}{(1-x)} = \frac{5R}{2} + \frac{R}{2} = 3R$

13. C

Sol. We know $E \propto \sqrt{P_{av}}$

$\frac{E_2}{E_1} = \sqrt{\frac{P_2}{P_1}} = \sqrt{\frac{100}{200}} = \frac{1}{\sqrt{2}}$

$\Rightarrow \qquad E_2 = \frac{E}{\sqrt{2}}$

14. D

Sol. From Conservation of momentum



$mv + 0 = mv_1 + 2mv_2$

$v = v_1 + 2v_2$ …(i)

coefficient of restitution;

$$e = \frac{v_2 - v_1}{v}$$

$v_2 - v_1 = v$ …(ii)

from (i) and (ii);

$$v_2 = \frac{2v}{3},\ v_1 = -\frac{v}{3}$$

$$\frac{\lambda_A}{\lambda_B} = \frac{2mv_2}{mv_1} = \frac{2v_2}{v_1} = 4$$

15. B

Sol. When $y = \frac{\beta}{4} = \frac{D\lambda}{4d}$

Path difference, $\Delta r = d \sin\theta = d\left(\frac{y}{D}\right) = d\left(\frac{\lambda}{4d}\right) = \frac{\lambda}{4}$ (for small 'θ')

$\therefore$ Phase difference, $\delta = \frac{2\pi}{\lambda} \times \Delta r = \frac{2\pi}{\lambda} \times \frac{\lambda}{4} = \frac{\pi}{2}$

$$I = I_0 \cos^2\left(\frac{\delta}{2}\right) = I_0 \cos^2\left(\frac{\pi}{4}\right) = \frac{I_0}{2}$$

16. D

Sol. $A\left(\frac{2\ell}{3}\right)(2\rho)g + A\left(\frac{\ell}{3}\right)\rho g = A\ell\rho_0 g$

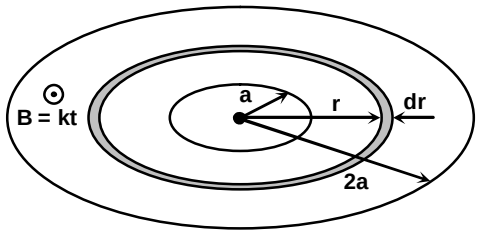
$$\frac{4\rho}{3} + \frac{\rho}{3} = \rho_0$$

$$\Rightarrow \rho_0 = \frac{5\rho}{3}$$

Specific gravity of the material of cylinder = $\rho_0/\rho = 5/3$

17. A

Sol. The emf induced in a thin circular ring of radius 'r' is



$$\varepsilon = \left|\frac{d\phi}{dt}\right|$$

$$\varepsilon = \frac{d}{dt}\left(kt\pi r^2\right) = k\pi r^2$$

$$\int dI = \int_a^{2a} \frac{k\pi r^2 h dr}{\rho 2\pi r}$$

$$I = \frac{kh}{2\rho}\left[\frac{r^2}{2}\right]_a^{2a} = \frac{3kha^2}{4\rho}$$

18. B

Sol. Shift, $dS = \left(1 - \frac{1}{\mu}\right)dx$

$$\int dS = \int_0^t \left(1 - \frac{t}{\mu_0(t+x)}\right) dx$$

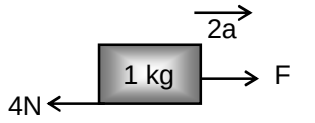
$\Rightarrow t - t\left[\ell n(t+x)\right]_0^t = t - t\ell n2 = t(1 - \ell n2)$

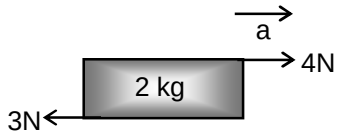
19. A

Sol. $F - 4 = 2a$ …(i)

$4 - 3 = 2a$ …(ii)

$\therefore F = 5N$





20. C

Sol. $I_C R_L + V_{CE} = V_{CC}$

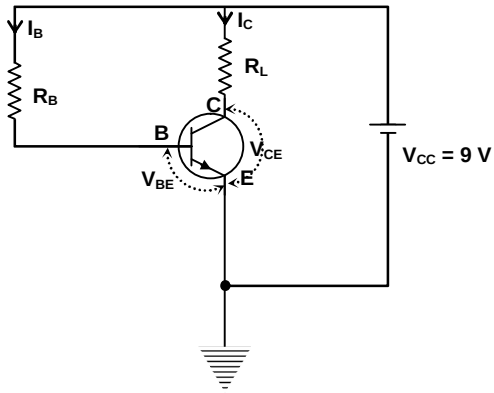
$I_C R_L = V_{CC} - V_{CE} = 9 - 5 = 4$ V

$\therefore I_C = \frac{4}{R_L} = \frac{4}{10^3} = 4 \times 10^{-3}$ A

$I_B = \frac{I_C}{\beta} = 4 \times 10^{-5}$ A

$V_{CC} = I_B R_B + V_{BE}$

$R_B = \frac{V_{CC} - V_{BE}}{I_B} = \frac{9 - 1.4}{4 \times 10^{-5}} = 1.9 \times 10^5 \Omega = 190\ k\Omega$

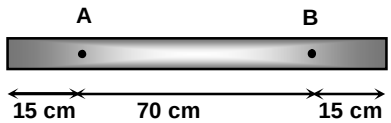


SECTION – B

21. 4

Sol. Velocity of longitudinal wave in a solid rod is

$v = \sqrt{\frac{Y}{\rho}} = \sqrt{\frac{1.6 \times 10^{11}}{2500}} = \frac{4}{5} \times 10^4 = 8000$ m/s



Now, $n_1 : n_2 : n_3 = \ell_1 : \ell_2 : \ell_3$

$= 15 : 70 : 15$

$= 1.5 : 7 : 1.5$

No. of loops formed in the fundamental mode of vibration

$= 1.5 + 7 + 1.5 = 10$ loops

$\therefore 10(\lambda/2) = \ell$

$\Rightarrow \lambda = (2\ell)/10 = 0.2$ m

$\therefore$ fundamental frequency, $f = \frac{v}{\lambda} = \frac{8 \times 10^3}{0.2} = 40$ kHz

22. 1

Sol. $a = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) g$

$$\frac{\Delta a}{a}=\frac{\Delta g}{g}+\frac{\Delta(m_1+m_2)}{m_1-m_2}+\frac{\Delta(m_1+m_2)}{m_1+m_2}$$

$$\frac{\Delta a}{a}=\frac{0.01}{9.80}+\frac{0.4}{60}+\frac{0.4}{147}=1\%$$

SECTION – C

23. 00002.82

Sol. $\omega = 2\pi n = 2\pi\times\frac{1}{4\pi}=0.5$ rad/s

$$\frac{1}{2}mv_r^2=\left(\frac{3}{2}\frac{KQq}{R}-\frac{KQq}{R}\right)+\int_0^R m\omega^2 r dr$$

$$\frac{1}{2}mv_r^2=\frac{KQq}{2R}+\frac{m\omega^2R^2}{2}\Rightarrow v_r^2=\frac{kQq}{R}+\omega^2R^2$$

$$v_r^2=\frac{9\times10^9\times8}{2}\times\frac{1}{6}\times10^{-9}+1=7\Rightarrow v_r=\sqrt{7}\text{ m/s}$$

$$v_t=\omega R=1\text{ m/s}$$

$$v=\sqrt{v_r^2+v_t^2}=\sqrt{7+1}=\sqrt{8}=2\sqrt{2}=2\times1.41=2.82\text{ m/s}$$

24. 00002.25

Sol. $v + f_e = 26$ cm

$\Rightarrow v = 26 - 2.5 = 23.5$ cm

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f_0}$$

$$1-\frac{v}{u}=\frac{v}{f_0}\Rightarrow\frac{v}{u}=1-\frac{v}{f_0}=1-\frac{23.5}{1}=-22.5\text{ cm}$$

Angular magnification, $m=\frac{v}{u}\frac{D}{f_e}=-22.5\times\frac{25}{2.5}=-225$

$|m| = 225$

$100x = 225 \Rightarrow x = 2.25$

25. 00042.50

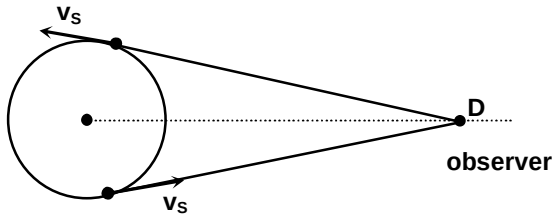
Sol. $$f_{max}=\left(\frac{v}{v-v_S}\right)f$$

$$f_{max}=\frac{340}{320}\times360\text{ Hz} = 382.5\text{ Hz}$$

$$f_{min}=\left(\frac{v}{v+v_S}\right)f$$

$$f_{min}=\frac{340}{360}\times360\text{ Hz} = 340.0\text{ Hz}$$

$\therefore\ f_{max}-f_{min}=42.50\text{ Hz}$



Chemistry

PART – II

SECTION – A

26. B
Sol. (A) The value of steric factor decreases as the complexity of reacting molecules increases.
(C) If chain termination is slower than chain branching and chain propagation, explosion occurs.
(D) N_2 is used to remove the excess energy as O_3 formed at lighter energy level can redissociate.

27. C
Sol. Stability order is
C > B > D > A

28. C
Sol. Due to highest electronegativity difference also due to bond order greater than 1 in BF_3 (only).

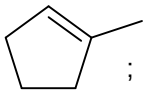
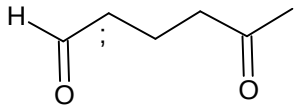
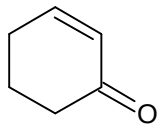
29. A
Sol. B & C are non aromatic as both are non planar.

30. D
Sol. Melting point of sodium (Na) is 371 K (approx), hence it is obtained in liquid state even at 850 K, the temperature of operation.

31. D
Sol. $\left[OH^-\right] = \sqrt{1.5 \times 10^{-8}}$ = 1.225 × 10^{-4}; pOH = 3.912;
Hence, pH = 10.088

32. A
Sol. (B) Duma's method, the organic compound is heated with CuO, not with Cu_2O.
(C) Fuming HNO_3 is used in Carius method.
(D) N, S, X (halogen) and phosphorus present in an organic compound are detected by Lassaigne's test.

33. D

34. B
Sol.
A =  ; B =  ; C = 

35. B
Sol. In A, C and D – S_N1 runs in parallel.

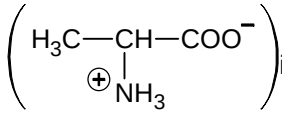
36. D
Sol. All are free radical reactions.

37. D
Sol. Carbon reduction is employed for extraction of Fe, Pb, Zn and Sn etc. generally.

38. B

Sol. Spin only magnetic moment of A, C and D are zero. 'C' cannot show ionization isomerism. While 'D' does not show geometrical isomerism.

39. C

Sol. By the processes defined in A, B and D, Alanine  is formed.

40. C

Sol. (A) $Al + dil.\ H_2SO_4 \longrightarrow Al_2(SO_4)_3 + H_2 \uparrow$

(B) $Al + AlCl_3 \xrightarrow{\text{High temperature}} AlCl$

(D) AlCl disproportionates below 800^oC to give $AlCl_3$ and Al.

41. D

Sol. Shapes of MX_2 (M = alkali earth metals) change from linear to bent from top to bottom.

42. C

Sol. $Be(OH)_2$ solution is amphoteric.

- pK_a of CH_3COOH is 4.76.
- pK_a of CH_3COOH is 3.76.

43. A

Sol. The order of corresponding equilibrium constants is : $K_4 < K_3 < K_2 < K_1$.

44. C

Sol. (A) Rain cloud – Aerosol

(B) Soapsuds – Foam

(D) Milk of magnesia – Sol

45. D

Sol. Chlorine water oxidizes Br^- to Br_2 and I^- to I_2.
Also, Br_2 water oxidize I^- to I_2.

SECTION – B

46. 8

Sol. For the given reaction : n = 2.

$E^o_{cell} = 0.24$

Hence, $E^o = \frac{0.06}{2}\log K_{eq}$

$\frac{0.24 \times 2}{0.06} = \log K_{eq}$

$K_{eq} = 10^8$

Hence, x = 8.

47. 3

Sol. $\underset{(\text{Glucose})}{C_6H_{12}O_6} \xrightarrow{HIO_4} 5HCOOH + HCHO;$

1 mole $\equiv$ 180 gm glucose give 30 gm HCHO
Hence, 18 gm glucose give 3 gm of HCHO.

SECTION – C

48. 00000.96

Sol. $K_P = 1.5 \times 1.5 = 2.25 \text{ atm}^2$

At new equilibrium:

$2.25 = P'_{H_2S} \times 2$

$P'_{H_2S} = 1.125$

$P_{total} = 3.125$

Hence, the desired ratio is $\frac{3}{3.125} = 0.96$

49. 00299.50

Sol. $H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(g);$

$\Delta H_f(H_2O, g) = \in_{H-H} + \frac{1}{2} \in_{O=O} -2 \times \in_{O-H}$

= - 299.50 (kJ/mole)

50. 00015.99

Sol. $\text{Density} = \frac{ZM}{N.a^3}; Z = 4(\text{fcc})$

M = 65 gm/mole
a = 300 pm = 3 × 10^{-8} cm.
Hence, density = 15.99 gm/cm^3.

Mathematics

PART – III

SECTION – A

51. A

Sol. $(\sim p \vee q) \vee (\sim q \vee r)$

52. C

Sol. The circles touch each other at (1, 1)

53. C

Sol. $x = \pm\sqrt{2}$

54. D

Sol. Use half angle formulae and Heron's formula

55. D

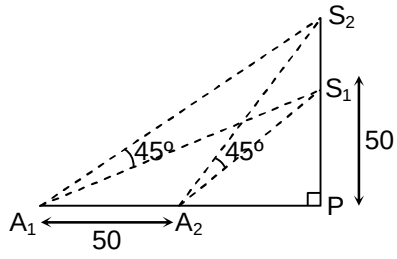
Sol. A_1, A_2, S_1 and S_2 are concyclic.

$\angle S_2A_1S_1 = \angle S_2A_2S_1 = 45°$ and $\angle S_1A_1A_2 = \angle S_1S_2A_2 = \alpha$

$\Rightarrow \angle A_1S_1A_2 = \angle A_1S_2A_2 = 45° - 2\alpha$

$$\frac{50}{\sin\left(\frac{\pi}{4} - 2\alpha\right)} = \frac{A_1S_1}{\sin\left(\frac{3\pi}{4} + \alpha\right)}$$, but $A_1S_1 = 50 \operatorname{cosec} \alpha$

$$\Rightarrow \tan\alpha = \frac{1}{3} \ \therefore \ S_1S_2 = \frac{A_1S_1}{\sin\left(\frac{\pi}{4} - \alpha\right)} \cdot \sin\frac{\pi}{4} = 250$$



56. B

Sol. Property, SP = Geometric mean of length of tangents

57. D

Sol. $\frac{x}{4} + \frac{y}{9} = 1$

58. B

Sol. Midpoint of segment between asymptotes is same as midpoint of chord

59. D

Sol. $x^2 - \frac{y^2}{66} = (\ell x + my)^2 \Rightarrow 1 - \ell^2 - \frac{1}{66} - m^2 = 0 \Rightarrow \ell^2 + m^2 = \frac{65}{66}$

60. B

Sol. $\int_{-3}^{3} f(x)dx \cdot \int_{0}^{1} f(x)dx < 0$, where $f(x) = ax^2 + bx + c$

61. D

Sol. z_1 lies on line joining 3 + 4i and 4 – 3i, which also contains the point 5 – 10i. So, expression is slope of perpendicular to this line

62. A

Sol. $\vec{c} = \lambda\left(\frac{\vec{a}}{|\vec{a}|} - + \frac{\vec{b}}{|\vec{b}|}\right)$

63. D

Sol. $A = \int_0^{\frac{1}{3}} \sqrt{12x}\, dx + \int_{\frac{1}{3}}^{1}\left(\frac{1}{x} - 1\right) dx$

64. D

Sol. $\lim_{n\to\infty} \frac{1}{n} \sum_{i=1}^{n-1} \frac{\frac{i}{n^2}}{1+\frac{i^4}{n^2}} = \int_0^1 \frac{x}{1+x^4} = \frac{1}{2}\tan^{-1} x^2\Big|_0^1 = \frac{\pi}{8}$

65. C

Sol. $\left(\frac{z+1}{z-1}\right)^{2n} = -1$

66. A

Sol. $ST^2 = AS^2 + AT^2 - 2AS\cdot AT\cdot\cos\alpha$ and $AS\cdot AT = \frac{1}{2}bc$

67. B

Sol. $t_n = \frac{1}{(n+1)(n+2)(n+3)} + \frac{1}{n(n+1)(n+2)(n+3)}$

68. B

Sol. $A = -1, B = 1$

69. A

Sol. $f(x) = x^2 (\tan x - \cos x)$

70. C

Sol. $xye^{2x} = \frac{xe^{3x}}{3} - \frac{e^{3x}}{9} + \frac{1}{9}$

SECTION – B

71. 6

Sol. $4!\cdot(2!\cdot 3 + 2 \times 2 + 3) = 312$

72. 9

Sol. $\sigma^2 = \frac{20\times 1 + 40\times 2^2}{60} = 3$ (Because mean of both samples is the same)

SECTION – C

73. 00001.50

Sol. $\left.\begin{array}{l} f\left(x^{\frac{1}{3}}\right) > 27x > f\left(x^{\frac{1}{3}} - 1\right) \\ f\left(\frac{x^{\frac{1}{3}}}{3}\right) > x > f\left(\frac{x^{\frac{1}{3}}}{3} - 1\right) \end{array}\right\}$ for large enough x, now use Sandwich Theorem.

74. 00004.50

Sol. Plane is $-x + z = 1 \Rightarrow d = \frac{3}{\sqrt{2}}$

75. 00000.50

Sol. p = 0.5 as there is a bijection between the sequences where 100th passenger sits in his assigned seat and the sequences where he doesn't.